

Aman Kumar Bhamboo

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Skills

Programming Languages: Python, SQL

Python Libraries: NumPy, Pandas, Seaborn, Matplotlib, Scikit-learn, Albumentations

Machine Learning: Data Preprocessing, Model Training & Evaluation, Feature Engineering, Classification & Regression Algorithms, Monte Carlo Dropout

Other Skills: Research & Literature Review, Prompt Engineering, Technical Writing

Tools & Platforms: Jupyter Notebook, Google Colab, Git, GitHub, Kaggle (Contributor), Entry-level Linux

Education

Bachelor of Technology in Artificial Intelligence and Machine Learning Oct 2022 – Jun 2026
Vellore Institute of Technology – Sehore, Madhya Pradesh CGPA: 7.38

Senior Secondary (XII), CBSE Apr 2020 – Aug 2021
Narendra Academy – Chirawa, Rajasthan Percentage: 87.40%

Secondary (X), CBSE Apr 2018 – Apr 2019
Dundlod Public School – Dundlod, Rajasthan Percentage: 94.20%

Experience

Researcher, Brain MRI Segmentation January 2025 – April 2025
• Published the paper "*Deep Learning-Based Approaches for Efficient Brain MRI Segmentation: A Comprehensive Analysis*" at IEEE STCR 2025.

- Conducted research on CNN-based models, with a focus on U-Net, for brain tumor segmentation using MRI data.
- Implemented a complete pipeline including preprocessing, augmentation, architecture design, training, and evaluation using Dice Loss and IoU metrics.

Independent Researcher, Multi-Modal Deep Learning for Brain Tumor Segmentation June 2025 – Present

- Extending previous work by incorporating multi-modal MRI inputs (T1, T2, FLAIR) for enhanced tumor segmentation performance.
- Preparing manuscript for submission to a peer-reviewed medical imaging journal (target: December 2025).

Projects

Brain Tumor Segmentation using U-Net with Monte Carlo Dropout [Github Link](#)

- Developed an ensemble U-Net architecture with Monte Carlo Dropout to segment brain tumors from MRI scans.
- Trained using Dice Loss and evaluated with Dice coefficient, achieving a test Dice score of 0.3189; analyzed performance gaps between ensemble models to guide future improvements.

PDF Summarizer – GPT-3.5 Turbo with LangChain and FAISS [Github Link](#)

- Built an interactive PDF summarization app using Streamlit, enabling users to upload documents and receive concise summaries via GPT-3.5 Turbo.
- Integrated LangChain, HuggingFace embeddings, and FAISS to preserve context through vector-based semantic retrieval and intelligent text chunking.

Certifications

- Completed Machine Learning Certification Course offered by Intershala. ([Link](#))
- Completed Python for Data Science, AI and Development offered by IBM. ([Link](#))
- Achieved contributor on Kaggle. ([Kaggle Link](#))